Transfer Learning for SAR Image Classification via Deep Joint Distribution Adaptation Networks

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***Abstract*— The problem of different characters of heterogeneous synthetic aperture radar (SAR) images leads to poor performances for transfer learning of SAR image classification. To address this issue, a semisupervised model named as deep joint distribution adaptation networks (DJDANs) is proposed for transfer learning from a source SAR image to a different but similar target SAR image, which aims to match the joint probability distributions between the source domain and target domain. In the proposed DJDAN, a marginal distribution adaptation network is developed to map features across the domains into an augmented common feature subspace, which aims to match the marginal probability distributions and unify the dimensions. Then, a conditional distribution adaptation network is proposed to transfer knowledge across the domains, which aims to reduce the discrepancies of the conditional probability distributions and enhance the effectiveness of feature representation. Moreover, one-versus-rest classification is utilized in the proposed framework, which aims to improve the discrimination between the inside and outside class. Experimental results demonstrate the effectiveness of the proposed deep networks.**

***Index Terms*— Deep neural networks (DNNs), domain adaptation (DA), image classification, synthetic aperture radar (SAR) image, transfer learning.**

I. Introduction

SYNTHETIC aperture radar (SAR) is an active microwave sensor that captures backscattering information of surface coverage in all weather conditions [1]. SAR images have been widely applied in many applications due to its capability of acquiring the characteristics of different objects. SAR image classification is to classify each pixel of the image to different categories, which is an important step in SAR image interpretation.

SAR images capture the polarimetric characteristics of the terrain objects, which can be utilized to distinguish different categories [2], [3]. The polarimetric properties of SAR data have been applied for classification, where typical polarimetric decomposition algorithms have been mostly adopted for unsupervised classification [4]–[6]. Cloude–Pottier decomposition, Freeman–Durden decomposition, and Yamaguchi decomposition have been developed for polarimetric SAR image classification. As for polarimetric SAR interferometry (PolInSAR) data, an improved four-component decomposition based on polarimetric interferometric similarity parameter was proposed for classification [7], which can solve the problem of volume component overestimation and improve classification accuracies. Moreover, an empirical volume scattering model using the repeat-pass PolInSAR coherence was proposed to solve the scattering mechanism ambiguity [8], which can improve the discrimination between forests and oriented built-up areas.

Machine learning-based classifiers have been widely adopted for SAR image classification in another branch. Support vector machine (SVM) [9], k-nearest neighbors [10], sparse representation classifier [11], and random forest [12] have been developed for SAR image classification. Recently, deep neural network (DNN) has been extensively applied for remote sensing interpretation [13]. DNN has the ability to learn high-level features that can reflect the intrinsical characters of the input data. As for classification tasks, DNN is able to enhance the discrimination of features, which contributes to the classification accuracy [14]. Typical deep models, including stacked autoencoders (SAEs), convolutional neural networks (CNNs), and deep belief networks (DBNs) have been developed for remote sensing image classification.

A series of work based on DNN has been proposed for SAR image classification. Chen *et al.* [15] combined multilayer projective dictionary pair learning and SAE for polarimetric SAR image classification, which achieves competitive classification accuracies. Zhao *et al.* [16] proposed a discriminant DBN to learn features from SAR images, which is combined with ensemble learning to produce better classification performance. Zhou *et al.* [17] constructed a four-layer CNN for polarimetric SAR image classification, where the 6-D real data converted from the covariance matrix are utilized as the input. Qin *et al.* [18] developed an adaptive boosting model based on restricted Boltzmann machines (RBMs), which yields a better result than the stacked RBM. Zhang *et al.* [19] proposed a complex-valued CNN for SAR image classification, which can adopt both amplitude and phase information of complex SAR images. Chen *et al*. [20] combined SAE and self-paced learning for classification, which gradually includes more difficult samples in the training process. In our previous work, a deep convolutional autoencoder (DCAE) [21] and a deep supervised and contractive neural network (DSCNN) [22] were

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proposed to learn effective features and conduct classification, which both yield superior accuracies for SAR image classification. Furthermore, an end-to-end deep model named as deep recurrent encoding neural networks (DRENNs) [23] was proposed to automatically extract contextual features from SAR images, which is able to enhance the discrimination of features and improve the classification accuracies.

All classification models mentioned above require numerous training samples to achieve competitive results. Generally, manual annotation is utilized to obtain training samples, which is a heavy task. In a real application, new data are acquired without labeled samples. To realize classification, transfer learning should be applied, where the new data should be classified by using the trained classifier on other data [24]. Compared with optical images, SAR works in the microwave frequency with emitting and receiving backscatter waves, and thus SAR images reflect different characters. The pretrained deep model on optical images cannot be applied for SAR images. At the same time, there are discrepancies of bands, resolutions, polarizations, or incidence angles for SAR data, i.e., heterogeneous SAR images. Therefore, during transfer learning for SAR image classification, the discrepancies of heterogeneous SAR data should be solved.

To deal with the above problem, domain adaptation (DA) methods have been recently developed for transfer learning, which can transfer the knowledge from a source domain to a different but related target domain [25]–[27]. The main idea of DA is to minimize the discrepancies of data across domains [28]. As for SAR images, different sensors, bands, polarizations, or incidence angles of the data can result in different domains. DA is able to overcome the discrepancies of different SAR data, which can match the characteristics between source data and target data. Hence, DA can involve samples from the source domain into the target domain, which tackles the issue of limited training samples in the target domain.

Various methods have been developed for DA, which can be mainly divided into four categories [28]: 1) semisupervised learning methods; 2) active learning approaches; 3) feature learning models; and 4) distribution matching and reweighting approaches. In semisupervised learning methods, labeled data of the target domain are not sufficient, and related data from the source domain are added into the training set to train the classifier. Transductive SVM (TSVM) models [29], graph-based models [30], and manifold regularization kernel machines [31] have been proposed for semisupervised DA. In active learning approaches, the training set is iteratively extended by selecting the most informative samples from the target domain [32]. SVM-based, kernel-based, and DNN-based [33] active learning methods have been developed for remote sensing tasks. In feature learning methods, a feature subspace is constructed to project features of the source and target domains together with minimizing the discrepancies [34]. In distribution matching and reweighting approaches, the probability distributions of the source and target domains are aligned in an unsupervised or a semisupervised way. Different methods for distribution matching have been developed to minimize the differences of the marginal distributions [35], conditional distributions [36], or both (joint distributions) [37].

Since different characters of heterogeneous SAR images lead to poor performances for transfer learning, a semisupervised model named as deep joint distribution adaptation networks (DJDANs) is proposed in this article. The proposed DJDAN aims to address the issue of transfer learning of heterogeneous SAR images, which can match the joint probability distributions of features of heterogeneous SAR images. Moreover, since categories of heterogeneous SAR images cannot be completely coincident, the issue of heterogeneous categories should be solved during transfer learning for SAR image classification. One-versus-rest classification is developed to solve heterogeneous categories by matching the class conditional probability distributions between source category and corresponding target category, where the similarities of heterogeneous categories are utilized to classify the target category. The proposed DNN is composed of a marginal distribution adaptation network, a conditional distribution adaptation network, and one-versus-rest classifiers. The main novel contributions of this article are summarized as follows.

1) A marginal distribution adaptation network is developed to match the marginal probability distributions across domains, which aims to solve the problems of different feature dimensions and different marginal distributions in heterogeneous SAR images. It is able to project the source and target features into an augmented common feature subspace, which efficiently learns domain invariant features.
2) A conditional distribution adaptation network is proposed to match the conditional probability distributions across domains, which aims to solve the shift of class conditional probability distributions in heterogeneous SAR images. It is able to transfer knowledge from source SAR image toward target SAR image, which reduces the discrepancies of intraclass feature between the source and target SAR images.
3) One-versus-rest classification is developed not only to improve the discrimination between the inside and outside class but also to conduct transfer learning among categories. Multiple one-versus-rest classifiers are utilized for relation learning of heterogeneous categories between the source domain and target domain. It is superior to the traditional multiclass classification method in transfer learning for SAR image classification.

The rest of this article is organized as follows. Section II presents the background of DA. The proposed DJDANs are described in Section III. Section IV reports and analyzes the experimental results. Conclusions are finally summarized in Section V.

II. Background

Traditional supervised learning assumes that the feature domain and the learning task of training and test data are both the same [38]. In transfer learning, the differences between training and test data are caused by the feature domain divergence or the learning task divergence, or both [39]. The purpose of transfer learning is to solve the domain shift between training and test data.

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A. DA

DA is identified as a specific subcategory of transfer learning, whose difference is induced by the feature domain divergences between the source domain $\mathcal{D}_s$ and the target domain $\mathcal{D}_t$. The learning tasks of DA in the source and target domains are the same, i.e., $\mathcal{T}_s = \mathcal{T}_t$. A feature domain $\mathcal{D}$ is composed of a feature space $\mathcal{X}$ and a probability distribution $P(\boldsymbol{X})$. Hence, the feature domain divergences can be split into different feature spaces $\mathcal{X}_s \neq \mathcal{X}_t$ and different probability distributions $P(\boldsymbol{X}^s) \neq P(\boldsymbol{X}^t)$. Based on the feature domain divergences, DA can be divided into homogeneous DA and heterogeneous DA.

As for homogeneous DA, the probability distributions between the source and target domains are nonuniform $[P(\boldsymbol{X}^s) \neq P(\boldsymbol{X}^t)]$, but the feature spaces across domains are equivalent with the same dimension ($\mathcal{X}_s = \mathcal{X}_t$). As for heterogeneous DA, the feature spaces between the source and target domains are unequal ($\mathcal{X}_s \neq \mathcal{X}_t$), where the dimensions across domains may be also unequal. To solve domain shift and transfer knowledge across domains, a basic assumption is that the source and target domains are directly relevant, which is satisfied in some cases. However, this assumption is not always valid in reality. Various intermediate bridges have been developed to enhance the correlation between the source and target domains [40].

B. Joint Distribution Adaptation (JDA)

In DA, the source and target data are performed with different probability distributions. To diminish the distribution divergences across domains, approaches for matching the marginal distribution or conditional distribution or both have been proposed. Long *et al.* [41] proposed the JDA model to jointly match the marginal distributions and conditional distributions across domains in a principled dimensional reduction procedure, which yields competitive results in heterogeneous SAR image classification.

In JDA, the labeled source domain is denoted as $\mathcal{D}_s = \{\mathcal{X}_s, P(\boldsymbol{x}^s)\}$, and the unlabeled target domain is denoted as $\mathcal{D}_t = \{\mathcal{X}_t, P(\boldsymbol{x}^t)\}$. The task of the source domain is represented as $\mathcal{T}_s = \{\mathcal{Y}_s, P(y^s|\boldsymbol{x}^s)\}$, and the task of the target domain is represented as $\mathcal{T}_t = \{\mathcal{Y}_t, P(y^t|\boldsymbol{x}^t)\}$. Assume that $\mathcal{X}_s = \mathcal{X}_t$, $\mathcal{Y}_s = \mathcal{Y}_t$, $P(\boldsymbol{x}^s) \neq P(\boldsymbol{x}^t)$, and $P(y^s|\boldsymbol{x}^s) \neq P(y^t|\boldsymbol{x}^t)$. The purpose of JDA is to reduce the distribution differences between $P(\boldsymbol{x}^s)$ and $P(\boldsymbol{x}^t)$, between $P(y^s|\boldsymbol{x}^s)$ and $P(y^t|\boldsymbol{x}^t)$ in a global framework. A feature space F is learned to adapt the joint probability distributions, which prefers to match the joint expectations of the features and labels across domains as follows:

$$\begin{aligned}\min_{F}\ & \|\mathbb{E}_{P(\boldsymbol{x}^s,y^s)}[F(\boldsymbol{x}^s), y^s] - \mathbb{E}_{P(\boldsymbol{x}^t,y^t)}[F(\boldsymbol{x}^t), y^t]\|^2 \\ \approx\ & \|\mathbb{E}_{P(\boldsymbol{x}^s)}[F(\boldsymbol{x}^s)] - \mathbb{E}_{P(\boldsymbol{x}^t)}[F(\boldsymbol{x}^t)]\|^2 \\ & + \|\mathbb{E}_{P(y^s|\boldsymbol{x}^s)}[y^s|F(\boldsymbol{x}^s)] - \mathbb{E}_{P(y^t|\boldsymbol{x}^t)}[y^t|F(\boldsymbol{x}^t)]\|^2 \quad (1)\end{aligned}$$

where $P(\boldsymbol{x}^s, y^s)$ and $P(\boldsymbol{x}^t, y^t)$ denote the joint probability distributions with respect to the source domain and the target domain, $\mathbb{E}[\cdot]$ represents the expectation function. Define a transformation matrix $\boldsymbol{A}$ to project data into the feature space F, and measure the differences of both marginal and conditional distributions by maximum mean discrepancy. To restrict the difference of the marginal distributions $P(\boldsymbol{x}^s)$ and $P(\boldsymbol{x}^t)$, the measure function can be written as

$$\left\| \frac{1}{N_s} \sum_{i=1}^{N_s} \boldsymbol{A}^T \boldsymbol{x}_i - \frac{1}{N_t} \sum_{j=1}^{N_t} \boldsymbol{A}^T \boldsymbol{x}_j \right\|^2 \quad (2)$$

where N_s and N_t denote the sample numbers with respect to the source domain and target domain. According to the sufficient statistics, class conditional distributions $P(\boldsymbol{x}^s|y^s)$ and $P(\boldsymbol{x}^t|y^t)$ are adopted to replace the posterior probability distributions $P(y^s|\boldsymbol{x}^s)$ and $P(y^t|\boldsymbol{x}^t)$, since the posterior probability distributions are quite complicated. Therefore, the measure function to restrict the difference of the conditional distributions is written as

$$\sum_{c=1}^{C} \left\| \frac{1}{N_s^c} \sum_{\boldsymbol{x}_i \in \mathcal{D}_s^c} \boldsymbol{A}^T \boldsymbol{x}_i - \frac{1}{N_t^c} \sum_{\boldsymbol{x}_j \in \mathcal{D}_t^c} \boldsymbol{A}^T \boldsymbol{x}_j \right\|^2 \quad (3)$$

where $\mathcal{D}_s^c$ and $\mathcal{D}_t^c$ stand for the samples belonging to the cth class in the source domain and target domain, respectively. N_s^c and N_t^c denote the sample numbers belonging to the cth class with respect to the source domain and target domain. C stands for the total number of class. Equations (2) and (3) can be jointly optimized to obtain the transformation matrix $\boldsymbol{A}$.

III. Methodology

In this article, DJDANs are proposed for SAR image classification, which aim to transfer learning from a source SAR image to a target SAR image. In the proposed DJDANs, a marginal distribution adaptation network is first developed to match the marginal probability distributions and solve the dimensional discrepancies. Then, a conditional distribution adaptation network is proposed to reduce the difference of the conditional probability distributions and enhance the discrimination of features. The overall framework consists of four modules: a contextual feature extraction model via our previous recurrent network [23], a marginal distribution adaptation network, a conditional distribution adaptation network, and a classification model, which is shown in Fig. 1.

A. Contextual Feature Extraction

The contextual feature has a great effect for the SAR image classification, and a series of contextual feature extraction models have been applied for SAR images. In our previous work [23], a recurrent network based on LSTM was proposed to extract contextual features from each SAR image patch automatically. Here, the size of each patch stands for the length of a square patch, and there is no overlap between adjacent image patches. It is proved that our recurrent network can discover contextual correlations among pixels and learn effective features from image patches, which outperforms other spatial feature extraction models. Therefore, a contextual feature extraction model via our previous recurrent network is utilized to extract features from both source and target SAR images.

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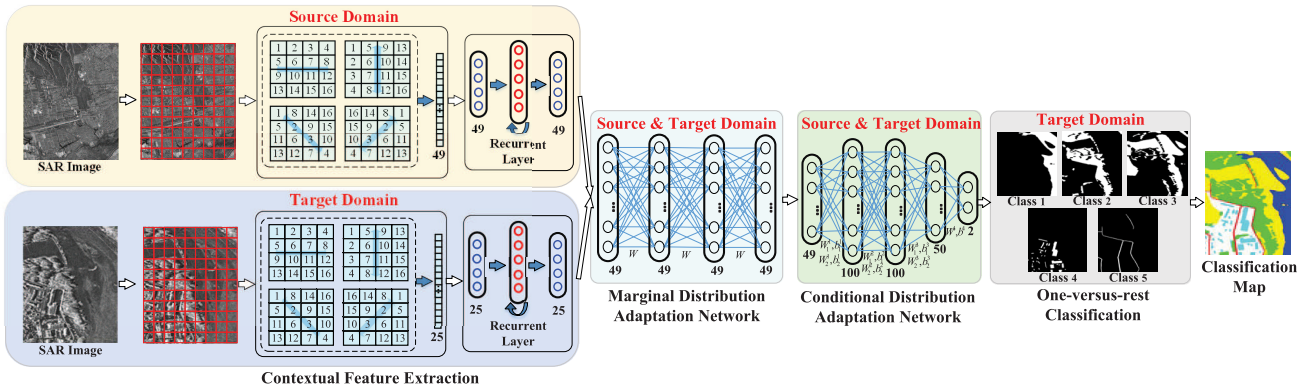


Fig. 1. Framework of DJDANs for SAR image classification.

B. Marginal Distribution Adaptation Network

In order to make the classifier trained on the source domain fitting for the target domain, features of the source domain and the target domain should be adapted. Since SAR data are captured from different sensors, bands, polarizations, and incidence angles, the transfer learning belongs to heterogeneous DA. Therefore, heterogeneous DA should be applied to reduce the discrepancies of SAR data across domains. As for SAR data of the source domain $\boldsymbol{X}^s$ and the target domain $\boldsymbol{X}^t$, the marginal probability distributions $P(\boldsymbol{X}^s)$ and $P(\boldsymbol{X}^t)$ are not consistent. A marginal distribution adaptation network is developed to learn a nonlinear subspace to match the marginal probability distributions, which is an improved model of marginalized stacked denoising autoencoder (mSDA) [35] by adding a marginal penalty term.

The proposed marginal distribution adaptation network is composed of multiple linear denoising encoders and nonlinear encoders, whose parameters can be simply estimated in closed form. A marginal penalty term is added into the objective function of linear denoising encoders to match the marginal probability distributions across domains. The source domain and the target domain are combined together $\mathcal{D} = \mathcal{D}_s \cup \mathcal{D}_t$, where the dimensions across domains are unified by adding zero to the low-dimensional features. Thus, the training sample of a combined set is represented as $\boldsymbol{x}_i$. Corrupt training samples by randomly setting each dimension to zero with probability p, and the mth corrupted version of $\boldsymbol{x}_i$ is denoted as $\boldsymbol{x}'_{i,m}$. Then, the linear denoising encoder is utilized to recover the original input from the corrupted data by linear mapping, i.e., $\hat{\boldsymbol{x}}_i = \boldsymbol{W}\boldsymbol{x}'_{i,m}$, where $\boldsymbol{W}$ denotes the linear mapping matrix. The objective function to obtain $\boldsymbol{W}$ is defined as follows:

$$J_M(\boldsymbol{W}) = \frac{1}{2NM}\left(\sum_{m=1}^{M}\sum_{i=1}^{N}\|\boldsymbol{x}_i - \boldsymbol{W}\boldsymbol{x}'_{i,m}\|_2^2 + \lambda\sum_{m=1}^{M}\left\|\frac{1}{N_s}\sum_{i=1}^{N_s}\boldsymbol{W}\boldsymbol{x}'_{i,m} - \frac{1}{N_t}\sum_{j=1}^{N_t}\boldsymbol{W}\boldsymbol{x}'_{j,m}\right\|_2^2\right) \quad (4)$$

where the first term is the squared reconstruction error, and the second term is the marginal penalty term. N_s and N_t denote the sample numbers with respect to the source domain and target domain, N represents the total number of training samples, i.e., $N = N_s + N_t$. M stands for the total version number of the corrupted data, and λ is the penalty parameter.

Define a matrix of all training samples as $\boldsymbol{X} = [\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_N]$, and the mth corrupted matrix as $\boldsymbol{X}'_m = [\boldsymbol{x}'_{1,m}, \boldsymbol{x}'_{2,m}, \ldots, \boldsymbol{x}'_{N,m}]$. Moreover, define a repeated matrix with M-times of $\boldsymbol{X}$ as $\hat{\boldsymbol{X}} = [\boldsymbol{X}, \boldsymbol{X}, \ldots, \boldsymbol{X}]$, and its corresponding corrupted version as $\boldsymbol{X}' = [\boldsymbol{X}'_1, \boldsymbol{X}'_2, \ldots, \boldsymbol{X}'_M]$. Thus, (4) can be rewritten as follows:

$$J_M(\boldsymbol{W}) = \frac{1}{2NM}\left[\|\hat{\boldsymbol{X}} - \boldsymbol{W}\boldsymbol{X}'\|_F^2 + \lambda \mathrm{tr}(\boldsymbol{W}\boldsymbol{X}'\boldsymbol{D}\boldsymbol{X}'^T\boldsymbol{W}^T)\right] \quad (5)$$

where $\boldsymbol{D}$ is a discrepancy index matrix, which is defined as

$$\boldsymbol{D} = \sum_{s\neq t}\boldsymbol{d}\boldsymbol{d}^T, \quad \text{where } d_i = \begin{cases} \dfrac{1}{N_s}, & \boldsymbol{x}_i \in \boldsymbol{X}_s \\ \dfrac{-1}{N_t}, & \boldsymbol{x}_i \in \boldsymbol{X}_t \end{cases} \quad (6)$$

where $\boldsymbol{d}$ is an $N \cdot M$-dimensional vector, whose element is denoted as d_i. Equation (5) has a closed-form solution as follows:

$$\boldsymbol{W} = (\hat{\boldsymbol{X}}\boldsymbol{X}'^T)(\boldsymbol{X}'\boldsymbol{X}'^T + \lambda\boldsymbol{X}'\boldsymbol{D}\boldsymbol{X}'^T)^{-1}. \quad (7)$$

When the version number M becomes larger, more corrupted versions are applied to obtain the linear mapping matrix $\boldsymbol{W}$. Based on the weak law of large numbers, the above solution converges to the expectation value when $M \to \infty$. Therefore, the linear mapping matrix $\boldsymbol{W}$ can be calculated as follows:

$$\boldsymbol{W} = \mathbb{E}[\hat{\boldsymbol{X}}\boldsymbol{X}'^T]\mathbb{E}[\boldsymbol{X}'\boldsymbol{X}'^T + \lambda\boldsymbol{X}'\boldsymbol{D}\boldsymbol{X}'^T]^{-1}. \quad (8)$$

Since the training samples are corrupted with probability p, the off-diagonal entries in the matrix $\boldsymbol{x}'_i\boldsymbol{x}'^T_i$ are uncorrupted with probability $(1-p)^2$. Set a vector $\boldsymbol{q} = [1-p, 1-p, \ldots, 1-p]$, in which $\boldsymbol{q}_a$ and $\boldsymbol{q}_b$ denote the uncorrupted probabilities with respect to the feature a and b. Set $\boldsymbol{Q} = \boldsymbol{X}\boldsymbol{X}^T$, and $\boldsymbol{Q}' = \boldsymbol{X}\boldsymbol{D}\boldsymbol{X}^T$. Hence, three parts of (8) can be represented as follows:

$$\mathbb{E}[\hat{\boldsymbol{X}}\boldsymbol{X}'^T]_{ab} = \boldsymbol{Q}_{ab}\boldsymbol{q}_b \quad (9)$$

$$\mathbb{E}[\boldsymbol{X}'\boldsymbol{X}'^T]_{ab} = \begin{cases} \boldsymbol{Q}_{ab}\boldsymbol{q}_a\boldsymbol{q}_b, & a \neq b \\ \boldsymbol{Q}_{ab}\boldsymbol{q}_a, & a = b \end{cases} \quad (10)$$

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$$\mathbb{E}[\boldsymbol{X}'\boldsymbol{D}\boldsymbol{X}'^{T}]_{ab} = \begin{cases} \boldsymbol{Q}'_{ab}\boldsymbol{q}_a\boldsymbol{q}_b, & a \neq b \\ \boldsymbol{Q}'_{ab}\boldsymbol{q}_a, & a = b. \end{cases} \tag{11}$$

Therefore, the matrix $\boldsymbol{W}$ can be calculated based on (8)–(11).

As we all know, DNN has a superb capability of feature learning due to the nonlinearity and the deep architecture. After the linear mapping with the closed-form solution, nonlinearity should be inserted into the network. Hence, a nonlinear encoder is connected with each linear denoising encoder for nonlinear representation. For the kth nonlinear encoder, the representation function is defined as follows:

$$\boldsymbol{h}^k = \tan h(\boldsymbol{h}^{k-1}) \tag{12}$$

where $\tan h(\cdot)$ is the nonlinear mapping function, and $\boldsymbol{h}^{k-1}$ denotes the input obtained from the kth linear denoising encoder. The output of the top nonlinear encoder stands for the features after matching the marginal probability distributions.

C. Conditional Distribution Adaptation Network

In classification tasks, minimizing the discrepancies of conditional probability distributions across domains is crucial for DA. The marginal distribution adaptation network cannot satisfy that the conditional distributions are also adapted. A conditional distribution adaptation network is proposed to transfer knowledge across domains, in which a conditional penalty term is proposed in the objective function of SAEs. Numerous training samples from the source domain and a small number of training samples from the target domain are utilized to reduce the shift of class conditional distributions $P(\boldsymbol{x}^s|y^s)$ and $P(\boldsymbol{x}^t|y^t)$.

The proposed conditional distribution adaptation network is connected after the marginal distribution adaptation network. Hence, output features from the marginal distribution adaptation network are imported into the conditional distribution adaptation network. Features of the source domain and the target domain are denoted as $\boldsymbol{h}^s_{i,k}$ and $\boldsymbol{h}^t_{i,k}$, respectively. Labels of the source domain and the target domain are denoted as $y(\boldsymbol{h}^s_{i,k})$ and $y(\boldsymbol{h}^t_{i,k})$, respectively. The encoding and decoding processes of the kth layer of the conditional distribution adaptation network can be written as

$$\boldsymbol{h}^*_{i,k} = f\left(\boldsymbol{W}^k_1\boldsymbol{h}^*_{i,k-1} + \boldsymbol{b}^k_1\right) \tag{13}$$

$$\hat{\boldsymbol{x}}^*_{i,k} = g\left(\boldsymbol{W}^k_2\boldsymbol{h}^*_{i,k} + \boldsymbol{b}^k_2\right) \tag{14}$$

where $f(\cdot)$ and $g(\cdot)$ denote the nonlinear functions with respect to the encoding and decoding processes; $\boldsymbol{W}^k_1$ and $\boldsymbol{W}^k_2$ stand for the weight matrices of the encoding and decoding processes, respectively; $\boldsymbol{b}^k_1$ and $\boldsymbol{b}^k_2$ denote the bias vectors of the encoding and decoding processes, respectively; $*$ represents s or t. Each autoencoder aims to minimize representation error between the encoding and decoding processes, whose objective function is defined as

$$J(\boldsymbol{\theta}^k) = \frac{1}{N}\sum_{i=1}^{N}\left\|\hat{\boldsymbol{x}}^*_{i,k} - \boldsymbol{h}^*_{i,k-1}\right\|_2^2 + \lambda'\left(\left\|\boldsymbol{W}^k_1\right\|_F^2 + \left\|\boldsymbol{W}^k_2\right\|_F^2\right) \tag{15}$$

where $\boldsymbol{\theta}^k = \{\boldsymbol{W}^k_1, \boldsymbol{W}^k_2, \boldsymbol{b}^k_1, \boldsymbol{b}^k_2\}$, the first term is the average reconstruction error between encoding and decoding, the second term is the weight penalty term, and λ' denotes the weight penalty parameter. The above function can be solved by the back propagation algorithm.

As for K layers of conditional distribution adaptation network, each layer is pretrained to optimize the parameters. After layer-by-layer training, the whole network is fine-tuned from the top layer to the bottom layer. A conditional penalty term is proposed in the fine-tuning objective function. The whole fine-tuning function is defined as follows:

$$J(\boldsymbol{\theta}) = \frac{1}{N}\sum_{k=1}^{K}\sum_{i=1}^{N}\left\|\boldsymbol{h}^*_{i,k} - \boldsymbol{h}^*_{i,k-1}\right\|_2^2 + \gamma\, J_C(\boldsymbol{\theta}) \tag{16}$$

where the first term is the total error of the encoding representation, and the second term is the conditional penalty term to reduce the differences of conditional probability distributions across domains, which is defined as

$$J_C(\boldsymbol{\theta}) = \frac{1}{C}\sum_{c=1}^{C}\left\|\frac{1}{N^c_s}\sum_{i=1}^{N^c_s}1\left\{y\left(\boldsymbol{h}^s_{i,k}\right)=c\right\}\boldsymbol{h}^s_{i,k} - \frac{1}{N^c_t}\sum_{j=1}^{N^c_t}1\left\{y\left(\boldsymbol{h}^t_{j,k}\right)=c\right\}\boldsymbol{h}^t_{j,k}\right\|_2^2 \tag{17}$$

where $\boldsymbol{h}^s_{i,k}$ and $\boldsymbol{h}^t_{j,k}$ denote the hidden representations with respect to the source sample and the target sample. The derivative of $J_C(\boldsymbol{\theta})$ is calculated as follows:

$$\frac{\partial J_C(\boldsymbol{\theta})}{\partial \boldsymbol{W}^k_1} = \frac{1}{C}\sum_{c=1}^{C}\left[\frac{1}{N^c_s}\sum_{i=1}^{N^c_s}1\left\{y\left(\boldsymbol{h}^s_{i,k}\right)=c\right\}\boldsymbol{v}_k(i,c)\boldsymbol{h}^s_{i,k-1} - \frac{1}{N^c_t}\sum_{j=1}^{N^c_t}1\left\{y\left(\boldsymbol{h}^t_{j,k}\right)=c\right\}\boldsymbol{v}_k(j,c)\boldsymbol{h}^t_{j,k-1}\right] \tag{18}$$

where the updating term $\boldsymbol{v}_k(i,c)$ has two situations when $k = K$ and $k \neq K$, which are defined as follows:

$$\boldsymbol{v}_K(i,c) = \left(\frac{1}{N^c_s}\sum_{i=1}^{N^c_s}1\left\{y\left(\boldsymbol{h}^s_{i,k}\right)=c\right\}\boldsymbol{h}^s_{i,k} - \frac{1}{N^c_t}\sum_{j=1}^{N^c_t}1\left\{y\left(\boldsymbol{h}^t_{j,k}\right)=c\right\}\boldsymbol{h}^t_{j,k}\right) f'\left(\boldsymbol{h}^s_{i,k}\right) \tag{19}$$

$$\boldsymbol{v}_k(i,c) = \boldsymbol{W}^k_1\boldsymbol{v}_{k+1}(i,c)f'\left(\boldsymbol{h}^s_{i,k}\right) \tag{20}$$

where $f'(\cdot)$ stands for the derivative of the nonlinear function $f(\cdot)$. Two situations of the updating term $\boldsymbol{v}_k(j,c)$ when $k = K$ and $k \neq K$ can be defined similarly.

The weights and biases of the overall conditional distribution adaptation network are fine-tuned based on the above functions. The hidden output of the top layer is the final feature representation after joint probability distribution adaptation.

D. Classification

SAR image classification is generally a multiclass classification task since various categories of land cover are contained in SAR images. In transfer learning for SAR image classification, the training samples of the target domain are not sufficient to

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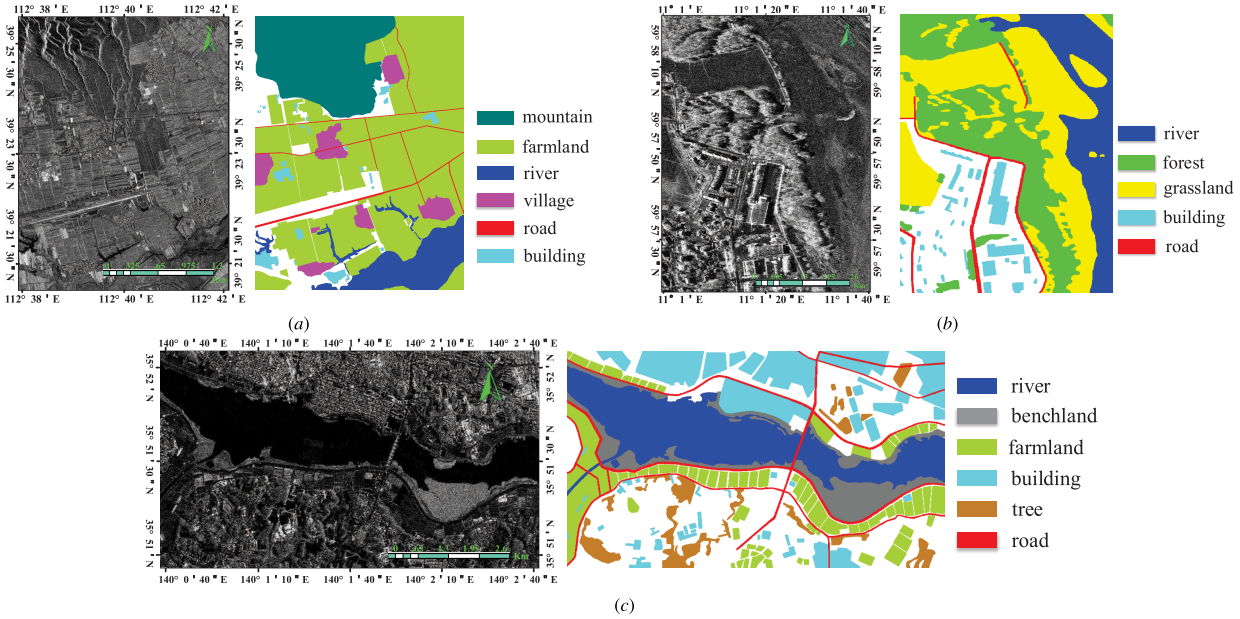


Fig. 2. SAR data adopted in the experiments. (a) Radarsat-2 data. (b) TerraSAR-X data. (c) ALOS-2 data.

train a multiclass classifier. To learn an easier classifier, one-versus-rest classification method is adopted in our framework, which can obtain the probabilities of each sample belonging or not belonging to every category. It aims to improve the discrimination between the inside and outside class. It is also able to solve the issue of heterogeneous categories across domains, where the similarities of heterogeneous categories are utilized to classify the target category.

Multiple one-versus-rest classifiers are conducted, where a softmax classifier is utilized in each classification. The predicted probability of the patch $\boldsymbol{x}_i$ belonging to the cth category is computed as follows:

$$p\left(y\left(\boldsymbol{h}_{i,K}^{t}\right)=c\right)=\frac{e^{\boldsymbol{W}_{c}^{K\,T}\boldsymbol{h}_{i,K}^{t}+b_{c}^{K}}}{\sum_{j=0}^{1}e^{\boldsymbol{W}_{j}^{K\,T}\boldsymbol{h}_{i,K}^{t}+b_{j}^{K}}} \tag{21}$$

where $\boldsymbol{W}_c^K$ and b_c^K denote the portion weight and bias of the softmax classifier, respectively. Here, $c = 1 or 0$, which stands for belonging or not belonging to the cth category. The binary classification label of the patch $\boldsymbol{x}_i$ is decided as follows:

$$L\left(\boldsymbol{h}_{i,K}^{t}\right)=\arg\max_{c=0or1}p\left(y\left(\boldsymbol{h}_{i,K}^{t}\right)=c\right). \tag{22}$$

Multiple binary classification results are utilized to decide the category of patches. The final label of each patch is determined with the maximal probability of the one-versus-rest classifiers, where probabilities of belonging to each category are used for comparison. The classification label of each patch stands for the labels of overall pixels in the patch. Finally, the classification labels of patches are transformed into a classification map based on the positions of patches.

IV. Experiments

A. Experimental Data

In the experiments, three SAR images are applied for transfer learning, which are Radarsat-2 data, TerraSAR-X data, and ALOS-2 data. These three data were obtained from different sensors with different bands, resolutions, polarizations, and incidence angles. The Radarsat-2 data reflect the area of Shuozhou, China. It contains six categories, which are mountain, farmland, river, village, road, and building. The TerraSAR-X data cover the region of Lillestroem, Norway. It includes five categories, which are river, forest, grassland, building, and road, respectively. The ALOS-2 data show the urban area of Tokyo, Japan. There are six categories in this data, which are river, benchland, farmland, building, tree, and road. The original SAR images and the corresponding ground-truth images are depicted in Fig. 2, where World Reference System-2 (WRS-2) is utilized for managing SAR images. More detailed descriptions about the data can be found in Table I. All the experiments in this article are executed in MATLAB2014, with Intel Xeon E5 1.6-GHz CPU and 32-GB memory.

TABLE I
Detailed Descriptions of Three Data

Data	Band	Polarization	Resolution	Image Size	Location
Radarsat-2	C	VV	1.5m	2000×1600	Shuozhou,China
TerraSAR-X	X	HH	0.38m	3580×2250	Lillestroem, Norway
ALOS-2	L	HH	3m	900×1600	Tokyo,Japan

To further verify the effectiveness of the proposed model, two data of the San Francisco Bay are utilized in the experiments, which are obtained by GaoFen-3 (GF-3) platform and AIRSAR platform, respectively. The subimage of GF-3 is applied as the source domain data, and the image of AIRSAR is applied as the target domain data. The Pauli color-coded images of two platforms and the ground truth image of AIRSAR are shown in Fig. 3. There are four categories in the two data, which are water, vegetation, low-density urban, and high-density urban [42].

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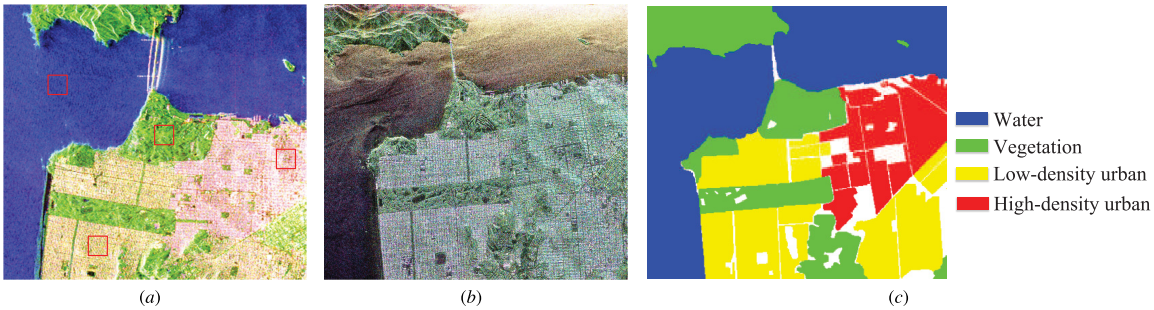


Fig. 3. San Francisco dataset. (a) GF-3 platform data. (b) AIRSAR platform data. (c) Ground truth image.

TABLE II
CORRESPONDING RELATIONS AMONG CATEGORIES FOR TRANSFER LEARNING

Radarsat-2→TerraSAR-X	Radarsat-2→ ALOS-2
river → river	river → river
mountain → forest	mountain → benchland
farmland → grassland	farmland → farmland
building→ building	building → building
road → road	village → tree
	road → road

To verify the effectiveness of the proposed DJDANs for transfer learning, the Radarsat-2 data are adopted as the source domain data, and the TerraSAR-X data and the ALOS-2 data are utilized as the target domain data. Each class accuracy, overall accuracy (OA), average accuracy (AA), and kappa coefficient (κ) are applied to evaluate the classification accuracies. In the proposed framework, our previous recurrent network [23] is applied to extract spatial features from the original SAR images. Based on our previous work, the patch size on the Radarsat-2 data is set to 7, and thus 49-D feature is extracted. As for the TerraSAR-X data, the patch size is set to 5, and 25-D feature is obtained. As for the ALOS-2 data, the patch size is set to 4, and 16-D feature is extracted. It is clear that the dimensions between the source and target domains are unequal, and the feature spaces are also different. Therefore, transfer learning from the Radarsat-2 data to the TerraSAR-X data or the ALOS-2 data is a heterogeneous DA.

Before DA, dimensions between the source and target domains should be unified by adding zero to the low-dimensional features. Moreover, categories between the source and target domains are not coincident, and thus heterogeneous transfer learning among categories should also be considered. In order to conduct classification, the corresponding relationships among categories are adopted, which is shown in Table II. When transfer learning occurs from the Radarsat-2 data to the TerraSAR-X data (Radarsat-2→ TerraSAR-X), forest and grassland are absent from the source domain, which are transferred from mountain and farmland, respectively. Similarly, benchland and tree of the ALOS-2 data are absent from the source domain, and they are, respectively, transferred from mountain and village during transfer learning from the Radarsat-2 data to the ALOS-2 data (Radarsat-2→ ALOS-2). To solve the issue of heterogeneous categories across domains, the similarities of corresponding categories between the source domain and target domain are utilized to classify the target category.

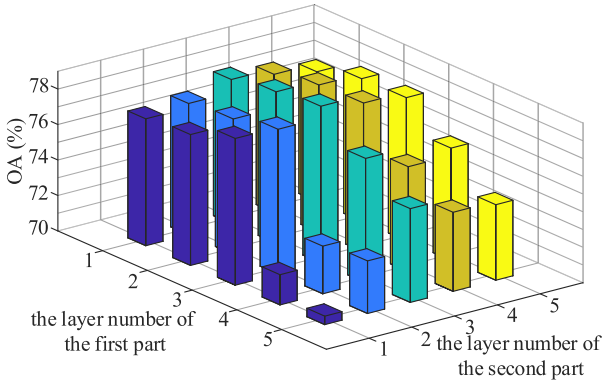


Fig. 4. Classification accuracies on the TerraSAR-X data with different hidden layer numbers of DJDANs.

B. Results on the TerraSAR-X Data

In the proposed DJDANs, numerous training samples from the source domain and only a small number of training samples from the target domain are utilized to train the deep network. In this section, the Radarsat-2 data are adopted as the source domain data for training, and a small percent of samples from the TerraSAR-X data are also selected for training. The classification performance of the proposed DJDAN is analyzed on the TerraSAR-X data.

The structure of the proposed DJDAN plays a crucial role on the performance of classification, which is analyzed first. DJDAN consists of two parts, which are a marginal distribution adaptation network and a conditional distribution adaptation network. The hidden layer numbers of these two parts are analyzed in order to obtain the optimal structure. The hidden unit number of the marginal distribution adaptation network is equal to the dimension of the input feature. The unit numbers of the front hidden layers of the conditional distribution adaptation network are set to 100, and the unit number of the last hidden layer is set to 50, which are set based on the experiment. The layer numbers of these two parts are discussed in Fig. 4.

It is clear that classification accuracy achieves the highest when the layer numbers of the two parts are both equal to 3. When fixing the layer number of the second part,

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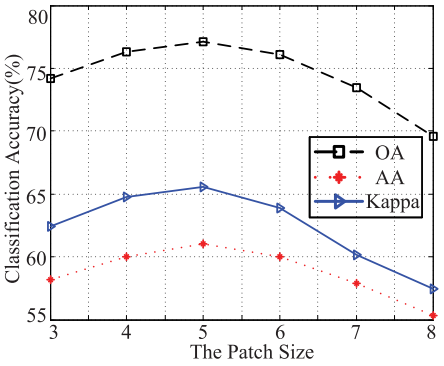


Fig. 5. Classification accuracies on the TerraSAR-X data with different patch sizes.

it is noted that classification accuracy increases first and then decreases with the rising of the layer number of the first part. A similar phenomenon happens when changing the layer number of the second part and fixing the layer number of the first part. It can be concluded that the layer number of the proposed DJDAN can be neither too small nor too large. When DJDAN is not deep enough, features between the source and target domains are not effectively matched. When DJDAN is too deeper, it may cause overfitting due to the limited training samples of the target domain. Therefore, DJDAN is composed of six hidden layers, including three layers of the marginal distribution adaptation network and three layers of the conditional distribution adaptation network.

In the experiments, the patch size is selected based on the classification performance. The analysis of the patch size is shown in Fig. 5, where the patch size is changed from 3 to 8. It is observed that classification accuracies achieve the optimal when the patch size is equal to 5. Classification accuracies increase first and then decrease with the changing of the patch size. When the patch size is small, the proposed network may not be able to capture effective spatial features. At the same time, the patch size cannot be too large due to the homogeneous property of the SAR image. Therefore, the patch size of the TerraSAR-X data is set to 5.

To yield the optimal classification accuracies, the parameters are analyzed in the experiments, respectively. Fig. 6 shows classification accuracies on the TerraSAR-X data with different parameters. The analysis of the parameter λ is shown in Fig. 6(a), where λ is changed from $1e-1$ to $1e-6$. It can be seen that classification accuracies increase first and then decrease with the changing of λ. When λ is equal to $1e-3$, the accuracies achieve the highest. The penalty parameters λ' and γ are analyzed in Fig. 6(b) and (c), respectively. Based on the results, penalty parameters λ, λ', and γ are set to $1e-3$, $5e-2$, and $5e-2$, respectively.

In the experiments, the Radarsat-2 image is utilized as the source data, which assumes that the ground-truth of the Radarsat-2 image is known. Hence, the whole samples from the Radarsat-2 data can be adopted as the training set. Classification results with different numbers of training samples on the source domain are analyzed in Fig. 7, where 500 to 4000 samples of each category are randomly selected from the Radarsat-2 data for training the deep network. Since some categories have fewer samples than 4000, all samples of these categories are utilized if more samples should be selected. In order to guarantee fair comparisons, 5% samples of each category from the TerraSAR-X data are randomly selected as the training set. According to Fig. 7, classification accuracies increase first and then decrease with the growing training number of the source domain. When the training number is equal to 2500, the accuracies achieve the highest. Therefore, 2500 training samples of each category from the source domain are enough to train the DJDAN network, and more samples can lead to the overfitting problem.

TABLE III
CLASSIFICATION COMPARISONS OF DIFFERENT SUBNETWORKS OF DJDAN ON THE TERRASAR-X DATA

	Marginal	Conditional	DJDAN (Marginal+Conditional)
OA	68.79	72.64	77.09
AA	50.14	54.20	61.01
Kappa	0.5019	0.5759	0.6559

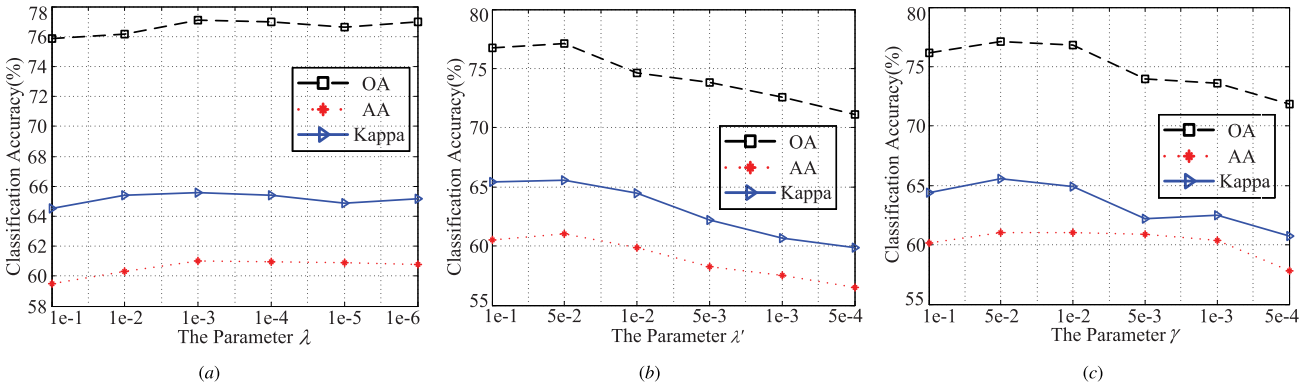
The proposed DJDAN is a semisupervised model with various labeled samples from the source domain and a small number of labeled samples from the target domain, which aims to decrease the demand of training samples from the target domain. The effect of training percent of the target domain on the classification performance is analyzed in Fig. 8, where 1%–10% samples of each category from the TerraSAR-X data are randomly selected as the training set, and 2500 training samples of each category from the Radarsat-2 data are also selected for training. It is remarkable that classification accuracies monotonically increase with the changing of training percent. When the training percent of the target domain is higher than 5%, the growth of the curve becomes slight. It can be illustrated that the proposed semisupervised classification model can produce superior results only with a small number of samples from the target domain.

To measure the classification performance, the confusion matrix is depicted in Fig. 9, where the row and column of the matrix denote true and predicted classes, respectively. The diagonal values stand for the accuracies of each category, and others reflect mis-probability. It is clear that the categories of forest, grassland, and building have superior accuracies, whose mis-probabilities are lower. The category of river is mis-classified into grassland with high probability. In addition, the category of road has quite unacceptable classification performance.

To analyze the effectiveness of each part of DJDAN, the classification results of each part are reported in Table III. DJDAN is proved to outperform the marginal distribution adaptation network and the conditional distribution adaptation network. It can be concluded that both the marginal distribution adaptation network and the conditional distribution adaptation network have contributions for transfer learning of SAR image classification. In addition, the conditional distribution adaptation network yields higher accuracies than the marginal distribution adaptation network.

Furthermore, the penalty terms of the proposed DJDAN are discussed in Table IV, where the weight term and the conditional term are analyzed. From the table, it is observed

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(*a*) (*b*) (*c*)

Fig. 6. Classification accuracies on the TerraSAR-X data with different parameters. (a) Parameter λ. (b) Parameter λ'. (c) Parameter γ.

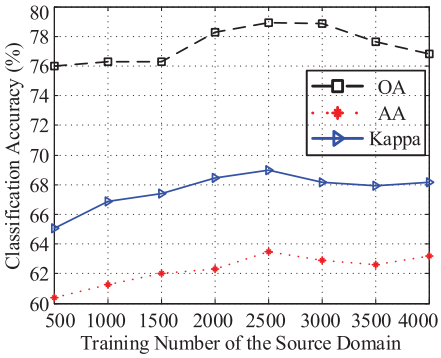


Fig. 7. Classification accuracies on the TerraSAR-X data with different training samples of the source domain.

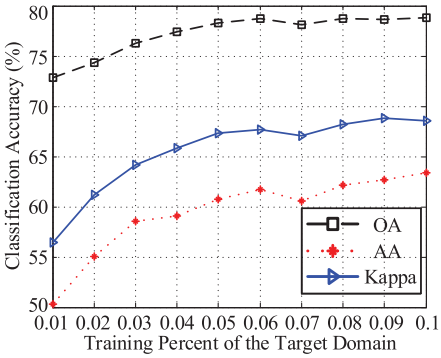


Fig. 8. Classification accuracies on the TerraSAR-X data with different training percents of the target domain.

TABLE IV

CLASSIFICATION COMPARISONS OF DJDAN WITH DIFFERENT PENALTY TERMS ON THE TERRASAR-X DATA

	weight penalty	conditional penalty	weight and conditional penalties
OA	60.41	76.27	77.09
AA	35.97	58.66	61.01
Kappa	0.4322	0.6365	0.6559

that the deep model with a conditional term produces better classification performance than that with the weight term. DJDAN with two penalty terms yields the highest accuracies,

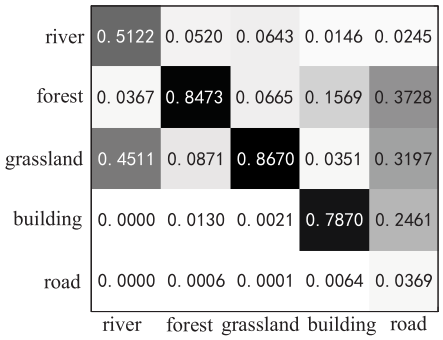


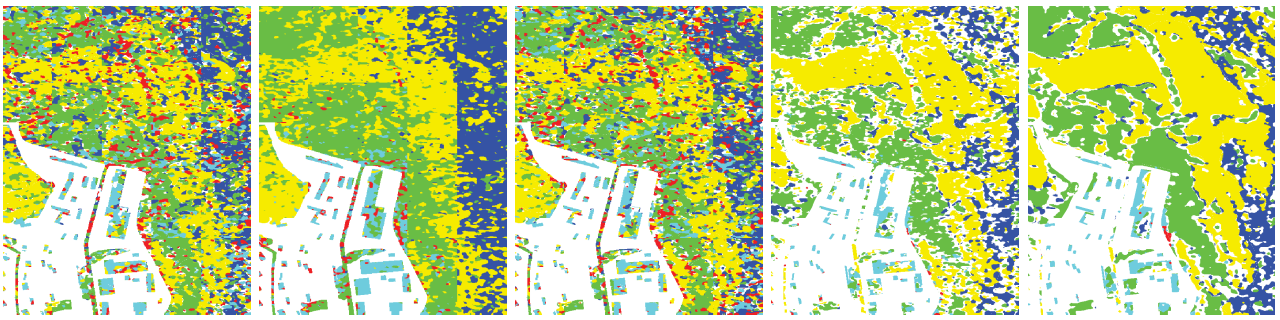
Fig. 9. Confusion matrix of the proposed DJDAN on the TerraSAR-X data.

which indicates that both the weight term and the conditional term have contributions to improve classification precision.

In order to verify the superiority of DJDAN for SAR image classification, the other four state-of-the-art DA methods are applied for comparison, which are transfer component analysis (TCA) [34], transfer kernel learning (TKL) [43], JDA [41], and mSDA [35]. Parameters of the compared approaches are set as the optimal to yield the best classification results. Classification comparisons on the TerraSAR-X data are reported in Table V, where each class accuracy, OA, AA, and κ are compared. In each model, 2500 training samples of each category from the Radarsat-2 data and 5% training samples from the TerraSAR-X data are randomly selected as the training set. Since some categories have fewer samples than 2500, all samples of these categories are adopted. Due to the issue of out of memory in the compared methods, the original data are divided into four subdata for operation. In addition, to verify the effectiveness of our one-versus-rest classification, traditional multiclass classification is applied for comparison in each model.

According to Table V, the proposed DJDAN yields much higher classification accuracies than other approaches. It is clear that DJDAN performs much better than TCA and TKL, which indicates that DJDAN has superior capability in transfer learning for SAR image classification than traditional DA methods. JDA, mSDA, and the proposed DJDAN are all developed for matching the probability distributions between the source and target domains. Compared with JDA and

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(a) (b) (c) (d) (e)

Fig. 10. Classification maps of each one-versus-rest model on the TerraSAR-X data. (a) TCA. (b) TKL. (c) JDA. (d) mSDA. (e) DJDAN.

TABLE V
COMPARISONS OF CLASSIFICATION RESULTS ON THE TERRASAR-X DATA

Target Domain			Source Domain		multiclass classification					one-versus-rest classification				
Class	Train	Test	Class	Train	TCA [34]	TKL [43]	JDA [41]	mSDA [35]	DJDAN	TCA [34]	TKL [43]	JDA [41]	mSDA [35]	**DJDAN**
river	2636	50081	river	2500	37.68	62.39	33.26	50.67	40.47	50.56	**67.45**	52.92	55.38	51.22
forest	3948	75006	mountain	2500	33.00	72.35	42.81	69.35	80.95	57.74	67.54	57.64	81.87	**84.73**
grassland	5119	97265	farmland	2500	33.19	66.19	41.17	73.13	79.67	61.04	59.75	61.60	81.22	**86.70**
building	724	13758	building	1689	54.66	55.32	25.41	61.16	47.74	67.82	66.62	66.32	72.87	**78.69**
road	397	7541	road	1998	29.43	16.49	11.46	7.55	0.05	35.20	22.74	**36.51**	7.88	3.69
OA					35.10	61.90	39.99	64.87	68.09	57.53	63.27	58.17	74.46	**77.09**
AA					37.59	54.55	30.82	52.37	49.78	54.47	56.82	54.60	59.85	**61.01**
κ					0.2146	0.5071	0.2601	0.4866	0.5238	0.4193	0.4776	0.4286	0.6194	**0.6559**

mSDA, the proposed DJDAN produces higher accuracies, which illustrates that DJDAN is able to better match the probability distributions across domains. Moreover, compared with multiclass classification, one-versus-rest classification performs better results, since it can train an easier classifier that improves the discrimination between the inside and outside class.

To further analyze the classification performances of each DA approach, the results of each approach with one-versus-rest classification are compared in Fig. 10. The proposed DJDAN shows the optimal visual effect on the TerraSAR-X data. The performance of mSDA is weaker, and the visual effect of the other three approaches is much worse. It is also observed that the proposed DJDAN yields superior visual effect on the categories of forest and grassland but the worse visual effect on the category of road. The performances of Fig. 10 are coincident with that of Table V. Therefore, DJDAN is able to produce effective classification results on the TerraSAR-X data.

The proposed DJDAN is an end-to-end deep network for SAR image classification, which is able to transfer learning automatically. To prove the superiority of the recurrent network, the traditional features are applied for comparisons, which are reported in Table VI. It can be observed that DJDAN with the contextual feature performs superior to DJDAN with traditional features. Moreover, classifiers for transfer learning are utilized to compare with the proposed network, which are TSVMs [44] and TradaBoost classifier [45]. Based on Table VI, the proposed DJDAN yields consistently higher accuracies with different features. This indicates that DJDAN has great contributions to improve the performance of SAR image classification, and the performance is derived from the deep model itself instead of the input. Therefore, the developed end-to-end deep network has superb performance in transfer learning of SAR image classification.

TABLE VI
CLASSIFICATION ACCURACIES OF DIFFERENT FEATURES ON THE TERRASAR-X DATA

		Pixels	Pixels-PCA	GLCM	Our Contextual Feature
TSVM[44]	OA	49.24	41.72	55.20	53.06
	AA	41.63	37.98	42.33	41.93
	Kappa	0.2870	0.1974	0.3182	0.2928
TradaBoost[45]	OA	58.33	50.90	56.88	54.76
	AA	40.69	39.06	42.15	41.86
	Kappa	0.3688	0.2035	0.3319	0.2976
DJDAN	OA	74.13	65.93	76.55	77.09
	AA	39.78	34.24	60.09	61.01
	Kappa	0.5688	0.4232	0.6487	0.6559

To further verify the performance of the proposed DJDAN, a CNN-based deep network is utilized for comparisons. Since optimization of the conditional distribution adaptation network is the same as the traditional deep network, it can be implemented using convolution networks. Thus, CNNs with a conditional penalty (CNN-Conditional) are adopted for comparisons. Moreover, segmentation-based methods are utilized for comparisons in the experiments. Fully convolutional networks (FCNs) [46] for semantic segmentation are applied for SAR image classification, whereas FCN-8s framework is applied in the experiments. Simple linear iterative clustering (SLIC) [47] is utilized for superpixel segmentation, which is combined with DJDAN for transfer learning. Due to the issue of out of memory in the compared methods, the original data are divided into two subdata for operation. Classification analyses of different deep models are reported in Table VII, and classification maps of different deep models are shown in Fig. 11.

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TABLE VII

CLASSIFICATION ANALYSES WITH DIFFERENT DEEP ARCHITECTURES ON THE TERRASAR-X DATA

	CNN-Conditional	FCN [46]	SLIC-DJDAN	DJDAN
OA	73.68	39.56	51.61	77.09
AA	60.90	37.42	35.54	61.01
Kappa	0.6160	0.1599	0.3078	0.6559

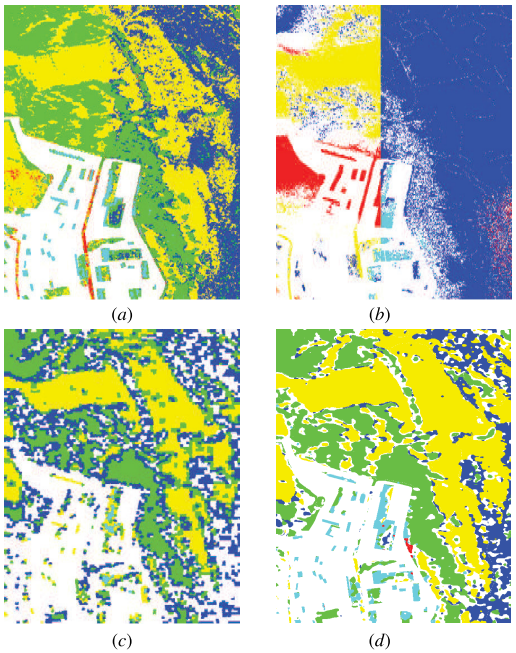


Fig. 11. Classification maps of different deep architectures on the TerraSAR-X data. (a) CNN-Conditional. (b) FCN. (c) SLIC-DJDAN. (d) DJDAN.

From the compared results of Table VII and Fig. 11, it is observed that the proposed deep network achieves the optimal classification results. The proposed DJDAN yields higher accuracies than the CNN-Conditional model. It is indicated that the proposed DJDAN has superior ability to match the joint probability distributions of features between the source and target data, which contributes to SAR image classification. Compared with segmentation-based methods, the proposed deep model performs much better. One possible explanation is that inherent speckle noise in the SAR image hampers the segmentation performance. For superpixel segmentation, some pixels belonging to different categories are falsely segmented into a superpixel, which may disrupt the spatial information and reduce the classification performance. The patch-based method takes pixels of spatial neighbors into consideration together, which can reduce the influence of speckle noise. Therefore, DJDAN produces better results than other deep models.

C. Results on the ALOS-2 Data

To broadly confirm the effectiveness of the proposed DJDAN, the ALOS-2 data is also utilized for transfer learning of SAR image classification. In this section, the Radarsat-2 data is adopted as the source domain data for training, and a small percent of samples from the ALOS-2 data are also selected as the training set.

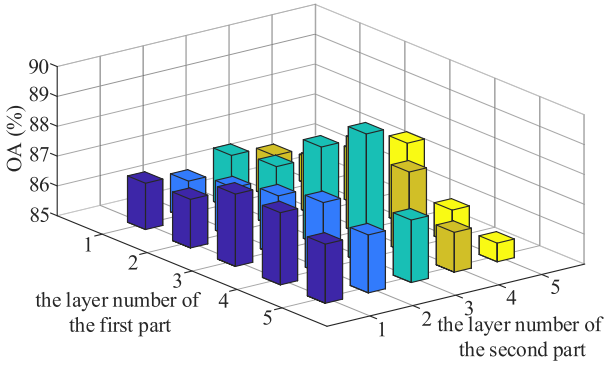


Fig. 12. Classification accuracies on the ALOS-2 data with different hidden layer numbers of DJDAN.

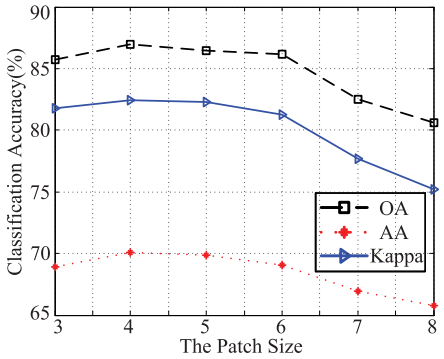


Fig. 13. Classification accuracies on the ALOS-2 data with different patch sizes.

According to the aforementioned experiments, the structure of DJDAN has a great effect on the classification accuracies. The hidden layer numbers of the two parts of DJDAN are also analyzed on the ALOS-2 data, which aims to establish the optimal architecture. Similar to the previous experiments, the hidden unit number of the first part is equal to the dimension of the input feature, the unit numbers of the front hidden layers of the second part are set to 100, and the unit number of the last hidden layer is set to 50. Classification accuracies with different hidden layer numbers of DJDAN are displayed in Fig. 12.

Based on the results in Fig. 12, DJDAN produces the highest accuracy when the layer number of the first part is equal to four and the layer number of the second part is equal to three. When fixing the layer number of one part and changing the layer number of the other part, classification accuracy increases first and then decreases with the rising of the layer number. It is illustrated that the layer number of the proposed DJDAN cannot be too small or too large. Therefore, DJDAN is composed of seven hidden layers on the ALOS-2 data, including four layers of the marginal distribution adaptation network and three layers of the conditional distribution adaptation network.

The patch size of the ALOS-2 data is also selected based on the classification performance, which is shown in Fig. 13. The accuracies achieve the optimal when the patch size is equal

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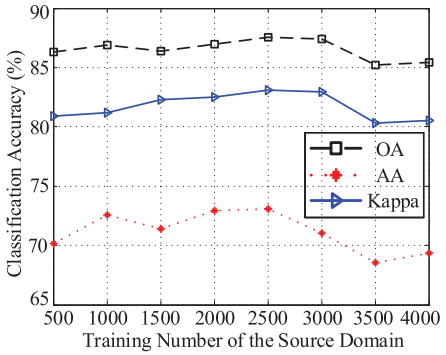


Fig. 14. Classification accuracies on the ALOS-2 data with different training samples of the source domain.

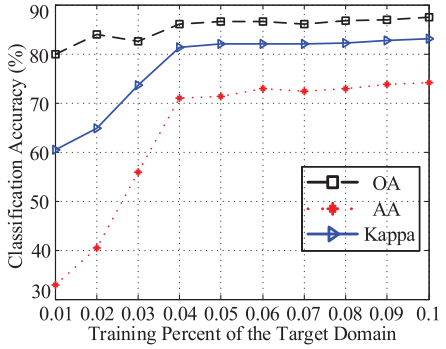


Fig. 15. Classification accuracies on the ALOS-2 data with different training percents of the target domain.

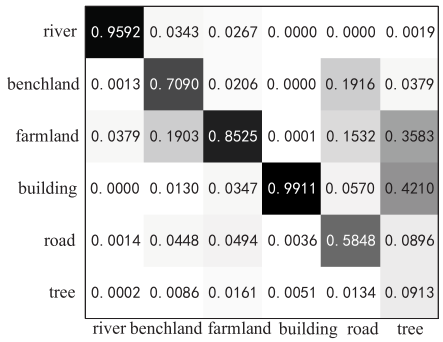


Fig. 16. Confusion matrix of the proposed DJDAN on the ALOS-2 data.

TABLE VIII
CLASSIFICATION COMPARISONS OF DIFFERENT SUBNETWORK OF DJDAN ON THE ALOS-2 DATA

	Marginal	Conditional	DJDAN (Marginal+Conditional)
OA	78.65	86.49	86.89
AA	57.87	57.77	69.80
Kappa	0.7128	0.8050	0.8230

TABLE IX
CLASSIFICATION COMPARISONS OF DJDAN WITH DIFFERENT PENALTY TERMS ON THE ALOS-2 DATA

	weight penalty	conditional penalty	weight and conditional penalties
OA	63.59	84.15	86.89
AA	17.26	63.53	69.80
Kappa	0.2436	0.7831	0.8230

to 4. The changing trends of curves in Figs. 5 and 13 are consistent. It can be concluded that the patch size has a great impact on the classification accuracies. Moreover, the patch size cannot be too large, since pixels in the larger patch have higher probabilities of belonging to different categories. Thus, the patch size of the ALOS-2 data is set to 4. From the results of Figs. 5 and 13, it is illustrated that homogeneous property of the SAR image affects choosing the patch size. Resolutions of the TerraSAR-X data and the ALOS-2 data are 0.38 and 3 m, respectively. That is to say, the TerraSAR-X data have higher spatial resolution than the ALOS-2 data, and thus the TerraSAR-X data have better homogeneous property. SAR image with better homogeneous property can select larger patch size. Therefore, we can choose the patch size based on the resolutions of SAR images, in which higher resolution SAR image can set larger patch size.

Classification performances on the ALOS-2 data with different training numbers of the source domain are analyzed in Fig. 14. Similar to the experiments on the TerraSAR-X data, 500 to 4000 samples of each category are randomly selected from the Radarsat-2 data as the training set, and 5% samples of each category from the ALOS-2 data are also randomly selected for training. It is noted from Fig. 14 that classification accuracies achieve the optimal with 2500 training samples from the source domain. Moreover, classification results increase first and then decrease with the rising of training numbers of the source domain.

The influence of training percents of the target domain on the ALOS-2 data is analyzed in Fig. 15, where 1%–10% samples of each category from the ALOS-2 data are randomly selected for training, and 2500 training samples of each category from the Radarsat-2 data are also selected as the training set. According to Fig. 15, classification accuracies monotonically increase with the growing number of training percent. The growth trend of the three curves becomes slight when the training percent of the target domain is higher than 5%. It can be concluded that our semisupervised classification model decreases the demand for training samples from the target domain.

To evaluate the classification result, Fig. 16 shows the confusion matrix on the ALOS-2 data. It is observed that the categories of river, benchland, farmland, and building show higher classification accuracies, but the categories of road and tree have worse classification performance. The category of the tree is mis-classified into farmland or building with high probability.

Classification accuracies of the two subnetworks are analyzed in Table VIII. The overall DJDAN yields higher accuracies than the two sub-networks on the ALOS-2 data. Both the marginal distribution adaptation network and the conditional distribution adaptation network have contributions to improve the performance of DJDAN. Moreover, the conditional distribution adaptation network yields higher accuracies than the marginal distribution adaptation network on the ALOS-2 data.

Table IX analyzes the effects of different penalty terms for classification performance. As we can see from the table, DJDAN with two penalty terms yields the highest accuracies,

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TABLE X

COMPARISONS OF CLASSIFICATION RESULTS ON THE ALOS-2 DATA

Target Domain			Source Domain		multiclass classification					one-versus-rest classification				
Class	Train	Test	Class	Train	TCA [34]	TKL [43]	JDA [41]	mSDA [35]	DJDAN	TCA [34]	TKL [43]	JDA [41]	mSDA [35]	**DJDAN**
river	856	16267	river	2500	79.11	83.62	79.24	85.11	89.87	66.22	77.26	85.13	86.18	**95.92**
benchland	373	7084	mountain	2500	37.62	25.15	44.76	39.81	55.18	30.84	17.84	54.39	48.39	**70.90**
farmland	625	11865	farmland	2500	63.44	66.68	62.82	66.92	73.09	67.57	62.52	69.37	83.72	**85.25**
building	743	14121	building	1689	79.53	87.33	85.09	93.54	95.26	69.25	68.49	82.11	97.84	**99.11**
tree	260	4945	village	2500	28.19	18.78	28.72	34.39	30.40	29.94	5.58	38.38	43.68	**58.48**
road	309	5879	road	1998	14.07	18.54	21.55	18.96	**25.36**	6.85	17.26	12.09	6.89	9.13
OA					53.66	64.54	68.31	69.49	74.54	56.79	61.18	70.78	80.41	**86.89**
AA					46.31	50.02	53.70	56.45	61.53	45.11	41.50	56.91	61.12	**69.80**
κ					0.4608	0.5465	0.5944	0.6132	0.6769	0.4575	0.4832	0.6229	0.7363	**0.8230**

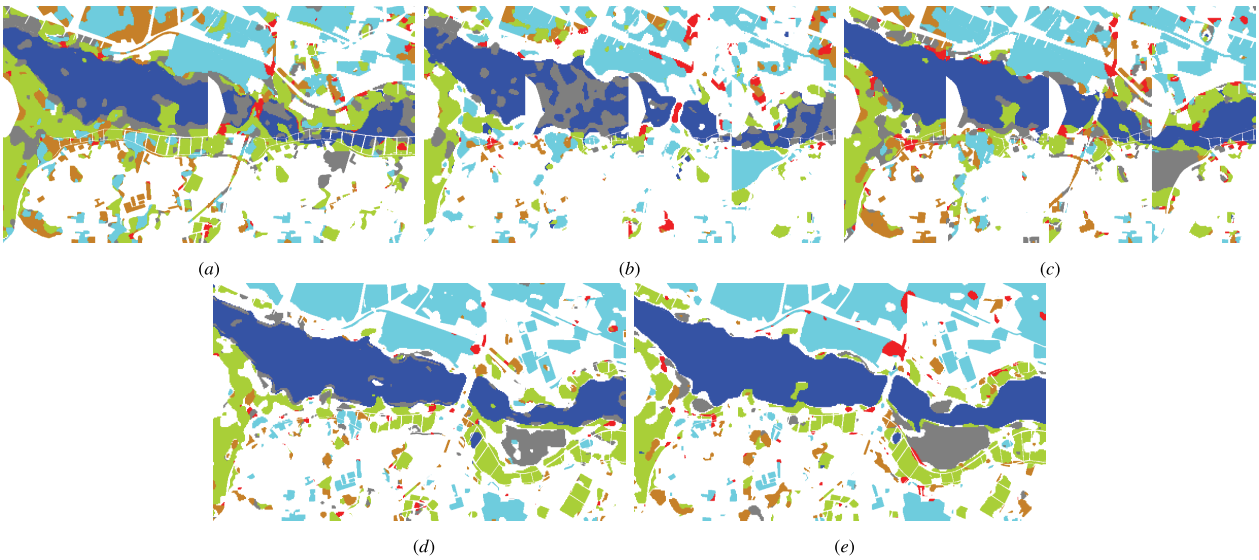


Fig. 17. Classification maps of each one-versus-rest model on the ALOS-2 data. (a) TCA. (b) TKL. (c) JDA. (d) mSDA. (e) DJDAN.

TABLE XI

CLASSIFICATION ACCURACIES OF DIFFERENT FEATURES ON THE ALOS-2 DATA

		Pixels	Pixels-PCA	GLCM	Our Contextual Feature
TSVM [44]	OA	50.19	48.26	66.42	73.45
	AA	23.07	20.97	31.18	45.20
	Kappa	0.1804	0.1584	0.4531	0.6334
TradaBoost [45]	OA	56.92	50.58	65.46	73.82
	AA	27.16	21.33	30.91	46.07
	Kappa	0.2149	0.1781	0.4412	0.6397
DJDAN	OA	78.38	65.22	84.49	86.89
	AA	46.20	19.99	63.58	69.80
	Kappa	0.6925	0.3605	0.7899	0.8230

TABLE XII

CLASSIFICATION ANALYSES WITH DIFFERENT DEEP ARCHITECTURES ON THE ALOS-2 DATA

	CNN-Conditional	FCN [46]	SLIC-DJDAN	DJDAN
OA	76.07	68.95	70.94	86.89
AA	61.74	51.53	54.39	69.80
Kappa	0.6923	0.5653	0.6155	0.8230

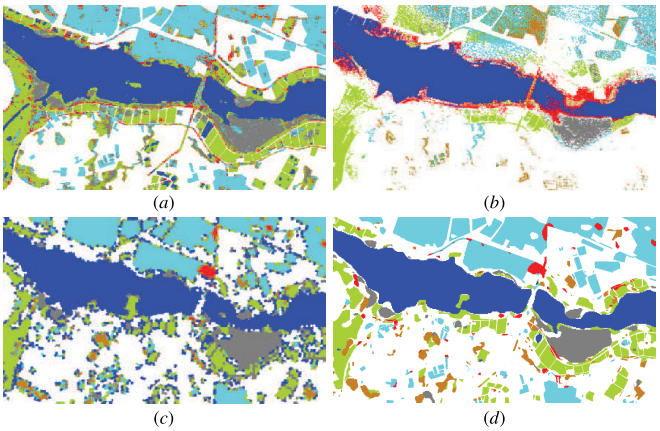


Fig. 18. Classification maps of different deep architectures on the ALOS-2 data. (a) CNN-Conditional. (b) FCN. (c) SLIC-DJDAN. (d) DJDAN.

and DJDAN with the conditional term outperforms DJDAN with the weight term. It can be concluded that both the weight term and the conditional term have contributions for SAR image transfer learning.

To verify the effectiveness of the proposed DJDAN model, TCA, TKL, JDA, and mSDA are applied for comparison. Parameters of the compared approaches are set to produce the best classification results on the ALOS-2 data. Furthermore, in order to prove the superiority of our one-versus-rest classification, traditional multiclass classification is utilized for comparison in each model. Classification results are compared in Table X, where 2500 training samples from the Radarsat-2 data and 5% training samples of each category from the ALOS-2 data are randomly selected as the training set.

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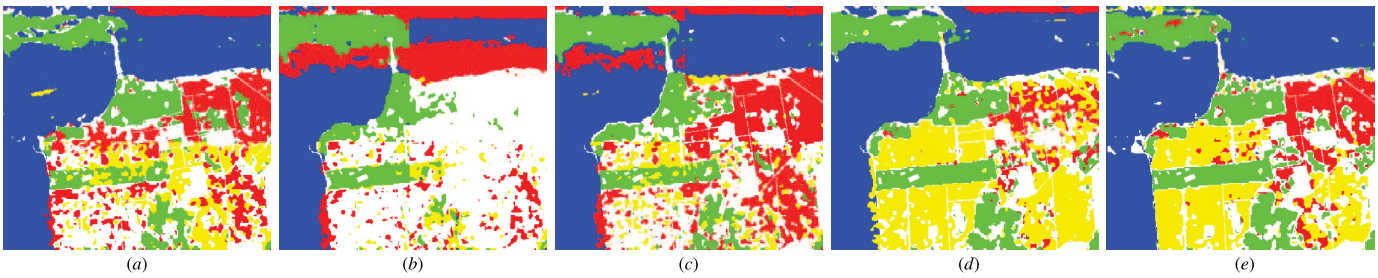


Fig. 19. Classification maps of each approach on the San Francisco data. (a) TCA. (b) TKL. (c) JDA. (d) mSDA. (e) DJDAN.

TABLE XIII
COMPARISONS OF CLASSIFICATION RESULTS ON THE SAN FRANCISCO DATA

Target Domain			Source Domain	TCA [34]	TKL [43]	JDA [41]	mSDA [35]	**DJDAN**
Class	Train	Test	Train					
water	3455	342010	2500	92.73	60.35	84.45	93.46	**99.13**
vegetation	1834	181560	2500	73.16	74.62	78.55	**89.22**	87.49
low-density urban	2072	205120	2500	42.68	22.72	33.09	**93.63**	86.13
high-density urban	1154	114192	2500	62.02	33.82	**89.23**	46.43	78.51
OA				76.20	60.94	75.35	87.38	**91.32**
AA				67.65	47.88	71.33	80.71	**87.81**
κ				0.6496	0.4222	0.6479	0.8172	**0.8743**

It is observed that DJDAN with one-versus-rest classification produces the best classification accuracies on the ALOS-2 data. Compared with TCA and TKL, the proposed DJDAN model yields much higher accuracies, which illustrates DJDAN outperforms traditional DA models for SAR image transfer learning. Compared with JDA and mSDA, our DJDAN obtains superior classification results, which indicates the proposed DJDAN has much better capability in distribution matching. In addition, classification results testify that one-versus-rest classification is superior to the traditional multi-class classification method, which improves the discrimination between the inside and outside class.

Furthermore, classification maps of each one-versus-rest approach are displayed in Fig. 17. It is worth noting that the proposed DJDAN model yields the optimal visual effect on the ALOS-2 data. The performances on the ALOS-2 data are coincident with those on the TerraSAR-X data. Therefore, the proposed DJDAN is able to produce competitive results for SAR image classification.

In the proposed DJDAN, the recurrent network is utilized to extract features from the original SAR image automatically, and thus DJDAN is an end-to-end deep network. In order to verify the performance of the recurrent network, Table XI reports the comparisons with the traditional features on the ALOS-2 data. DJDAN with the contextual feature yields higher accuracies than DJDAN with traditional features. In addition, related classifiers for transfer learning are compared with the proposed DJDAN model on the ALOS-2 data. The classification results are reported in Table XI. It is obvious that the accuracies of the proposed DJDAN are much higher than the other two classifiers in each case, which illustrates that DJDAN can improve the transfer learning performance of SAR images. DJDAN with the contextual feature yields the optimal accuracies, which also indicates the effectiveness of our contextual features. Therefore, our end-to-end DJDAN is able to produce superb classification results in SAR image transfer learning.

To further prove the performance of DJDAN, CNN-Conditional model, FCN model [46] and SLIC-DJDAN model are also compared on the ALOS-2 data. Classification comparisons of different deep models are reported in Table XII. It is seen that the proposed DJDAN achieves the best classification accuracies on the ALOS-2 data. It is concluded that the proposed DJDAN has superb ability to match the joint probability distributions of features between the source and target SAR data, which contributes to the SAR image classification.

Furthermore, CNN-based model and segmentation-based models are applied to verify the performance of the proposed network. Specifically, CNN with a conditional penalty (CNN-Conditional) and SLIC superpixel segmentation with DJDAN (SLIC-DJDAN) are applied for comparisons, which are shown in Table XII and Fig. 18. It is clear that our DJDAN yields better results than the other deep models. The proposed network has the superb capability to reduce the feature discrepancies and category discrepancies, which can contribute to the classification of heterogeneous SAR images. The patch-based method takes pixels of spatial neighbors into consideration together, which is able to reduce the influence of speckle noise in SAR images.

D. Results on the San Francisco Data

Two data of the San Francisco Bay are utilized for transfer learning of SAR image classification, which is obtained by GF-3 platform and AIRSAR platform, respectively. Samples are selected from GF-3 as the source domain data, which are from the four red squares in Fig. 3(a). Only 1% of samples from AIRSAR are also applied for training. The compared results of transfer learning are reported in Table XIII.

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It is obvious that the proposed DJDAN produces superior classification performance than other related methods. Compared with TCA and TKL, DJDAN performs higher accuracies, which illustrates DJDAN has a superior capability for SAR image classification than traditional DA methods. Compared with JDA and mSDA, the proposed DJDAN yields better results since it is able to match better the probability distributions across domains. The experimental results are consistent with the results of the aforementioned data, which indicates the proposed DJDAN has superb and stable performance for transfer learning of SAR images.

The visual effect of each approach is compared to the San Francisco data, which is shown in Fig. 19. It can be observed that DJDAN yields superior visual effect, where there is fewer misclassification between water and high-density urban in the proposed network. Therefore, DJDAN can produce effective classification of SAR images.

V. Conclusion

In this article, DJDANs are proposed for SAR image classification, which aim to match the joint probability distributions of features between the source and target SAR images. Two transfer learning experiments demonstrate that DJDAN is able to yield higher classification accuracies than other related methods. In the DJDAN model, a marginal distribution adaptation network is developed to match the marginal distributions and unify the dimensions between the source and target features, and a conditional distribution adaptation network is proposed to reduce the discrepancies of the conditional distributions and enhance the discrimination of features. Therefore, the proposed DJDAN has the capability to match the joint probability distributions between the source and target domains, which can improve the accuracies for SAR image classification. In addition, a one-versus-rest classification approach is developed in our framework, which is verified to perform better than the multiclass classification approach.

Since the patch-based method is utilized in DJDAN, it may lead to the block result. We prefer to develop segmentation based methods for SAR image classification in our future work.

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